

Task 6: Audit S3 Buckets for Public Access and Notify

Step 1: Create one S3 bucket with Disable Block Public Access

- Created an S3 bucket named **srikanth-s3-audit-public-test**.
- Disabled all Block Public Access settings for the bucket after creation.
- Modified the bucket permissions to make the bucket publicly accessible for testing the S3 public access audit.

srikanth-s3-audit-public-test [Info](#)

Objects | Metadata | Properties | **Permissions** | Metrics | Management | File systems - new | Access Points

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).
[View analyzer for us-east-1](#)

Block public access (bucket settings) [Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access
 Off
► Individual Block Public Access settings for this bucket

Bucket policy [Edit](#) [Delete](#)

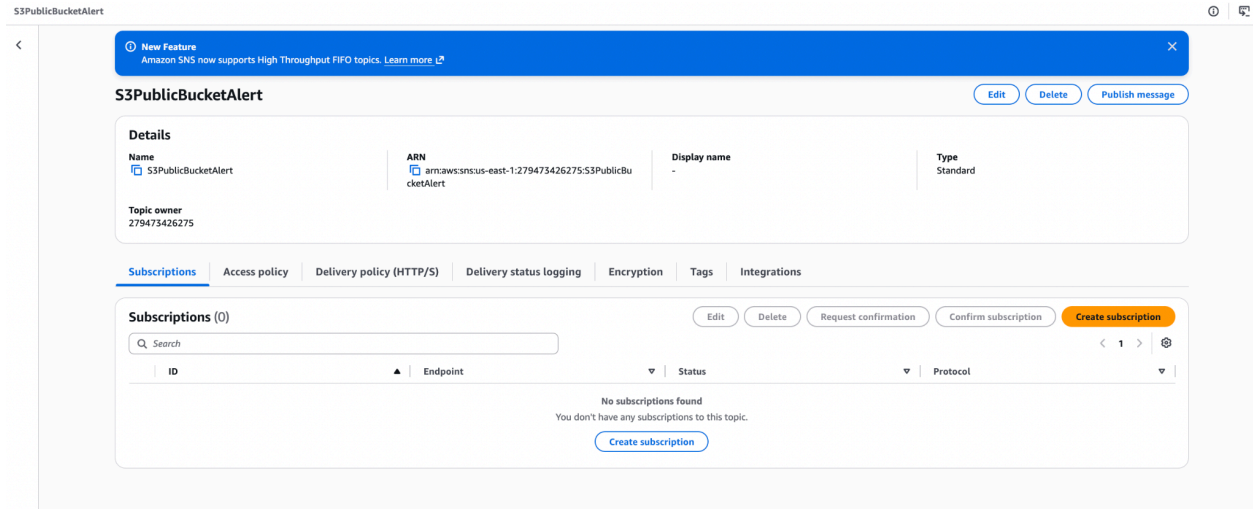
The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicRead",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::srikanth-s3-audit-public-test/*"
    }
  ]
}
```

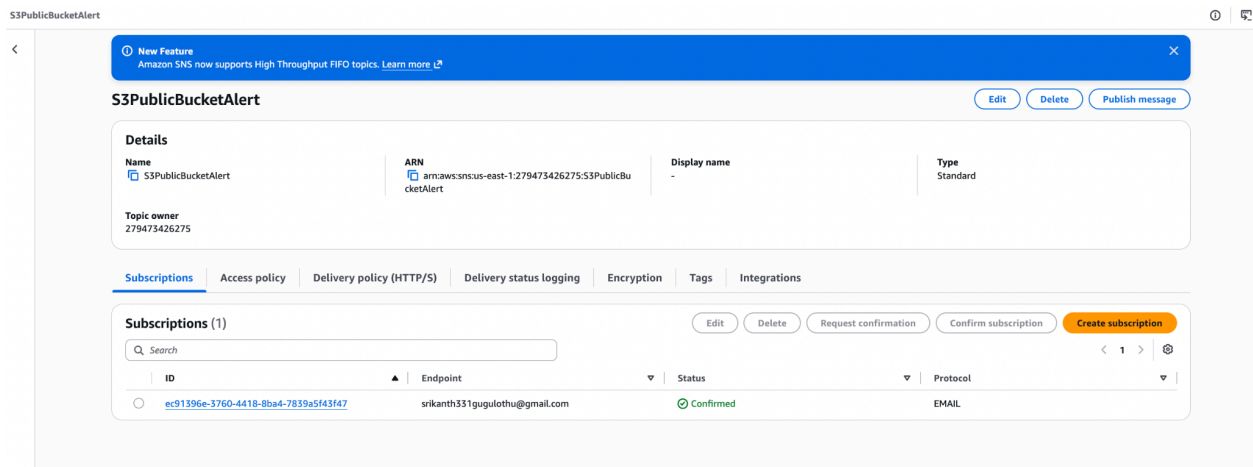
[Copy](#)

Step 2: Create an SNS Topic

Created an SNS topic with the name **S3PublicBucketAlert**



Created a Gmail email subscription for the **S3PublicBucketAlert** SNS topic



Step 3: Create IAM Role for Lambda

Create an IAM role with the following configuration:

- **Trusted Entity:** AWS Service
- **Use Case:** Lambda
- **Permissions Policy:** **AWSLambdaBasicExecutionRole**
- **Role Name:** **S3BucketCleanupRole**

S3BucketCleanupRoleInfoDelete

Allows Lambda functions to call AWS services on your behalf.

SummaryEdit

Creation date

July 23, 2026, 12:49 (UTC+05:30)

Last activity

-

ARN

arn:aws:iam::279473426275:role/S3BucketCleanupRole

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1)Info

SimulateL2

Remove

Add permissions

You can attach up to 10 managed policies.

Filter by Type

All types

< 1 >

Policy nameL2

Type

Attached entities

AWSLambdaBasicExecutionRole

AWS managed

3

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. Learn moreL2

Generate policy

No requests to generate a policy in the past 7 days.

Add Inline Policy to IAM user

Policy LambdaS3Audit created.

S3BucketCleanupRoleInfoDelete

Allows Lambda functions to call AWS services on your behalf.

SummaryEdit

Creation date

July 23, 2026, 12:49 (UTC+05:30)

Last activity

-

ARN

arn:aws:iam::279473426275:role/S3BucketCleanupRole

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (2)Info

SimulateL2

Remove

Add permissions

You can attach up to 10 managed policies.

Filter by Type

All types

< 1 >

Policy nameL2

Type

Attached entities

AWSLambdaBasicExecutionRole

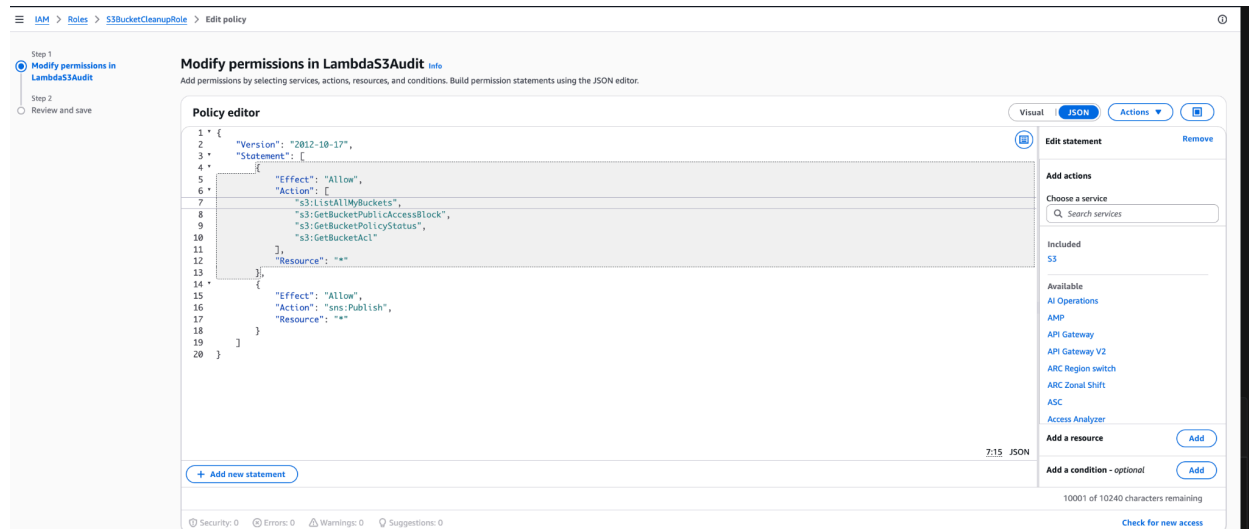
AWS managed

3

LambdaS3Audit

Customer inline

0



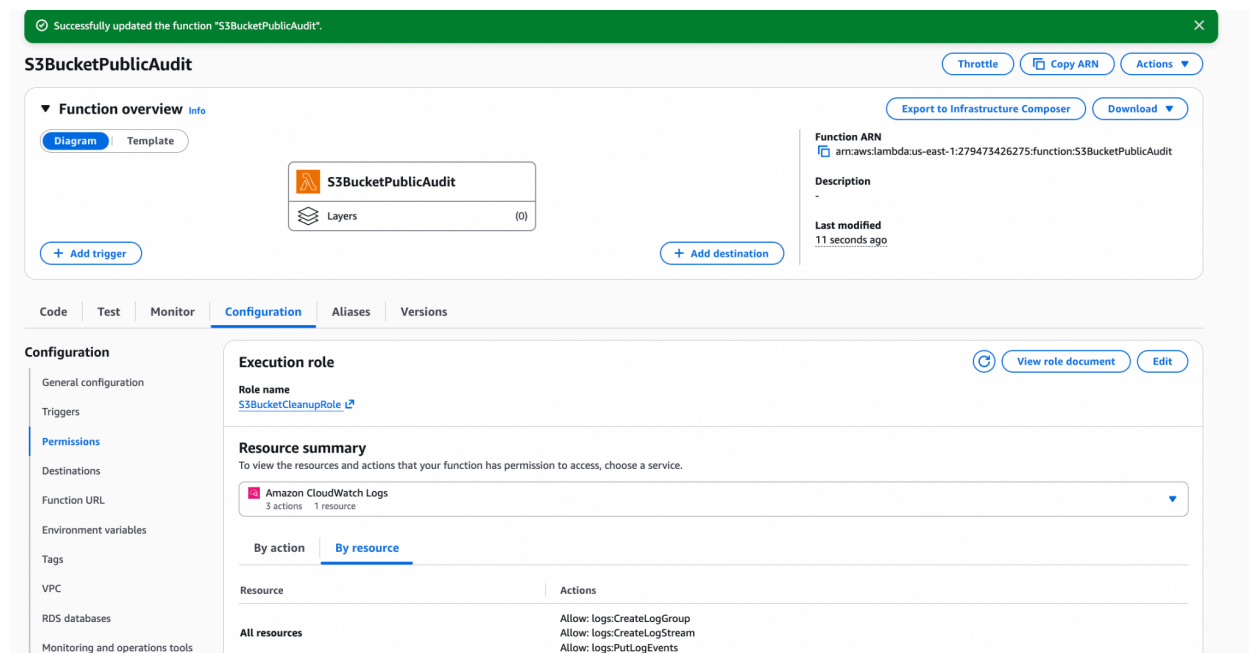
Step 4: Create Lambda Function

Create an AWS Lambda function with the following configuration:

Function Name: S3BucketPublicAudit

Runtime: Python 3.14

Execution Role: S3BucketCleanupRole



Changed the code and successfully deployed it

Step 5: Test Lambda

Create a test event to invoke the Lambda function with the following configuration:

- Event Name: **AuditTest**
- Event JSON:



Code
Test
Monitor
Configuration
Aliases
Versions

Test event
Info

CloudWatch Logs Live Tail
Save
Test

To invoke your function without saving an event, configure the JSON event, then choose Test.

Test event action

Create new event
Edit saved event

Invocation type

☒ Synchronous
Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

☐ Asynchronous
Enqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.

Event name

AuditTest

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings

☒ Private
This event is only available in the Lambda console and to the event creator. You can configure a total of 10. [Learn more](#)

☐ Shareable
This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional

Hello World

Event JSON

Format JSON

1 {
2 }

Run the test

Expected CloudWatch Logs but No public bucket found

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search
Clear
1m
30m
1h
12h
Custom
UTC timezone
Display

Timestamp	Message
	No older events at this moment. Retry
2026-07-23T11:10:47.637Z	INIT_START Runtime Version: python:3.14.v50 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:cb1b08fe2b02c2367290ef4fb2720dc1ee0568820fe174fe3df95631f7b85
2026-07-23T11:10:48.164Z	START RequestId: aac609eb-4225-435a-a9d9-96f1ec618f40 Version: \$LATEST
2026-07-23T11:10:49.588Z	No public buckets found.
2026-07-23T11:10:49.689Z	END RequestId: aac609eb-4225-435a-a9d9-96f1ec618f40
2026-07-23T11:10:49.689Z	REPORT RequestId: aac609eb-4225-435a-a9d9-96f1ec618f40 Duration: 1443.84 ms Billed Duration: 1968 ms Memory Size: 128 MB Max Memory Used: 98 MB Init Duration: 523.76 ms
2026-07-23T11:12:11.228Z	START RequestId: dca7ce47-e09c-4e39-a105-341b8cbe2c83 Version: \$LATEST
2026-07-23T11:12:12.588Z	No public buckets found.
2026-07-23T11:12:12.529Z	END RequestId: dca7ce47-e09c-4e39-a105-341b8cbe2c83
2026-07-23T11:12:12.529Z	REPORT RequestId: dca7ce47-e09c-4e39-a105-341b8cbe2c83 Duration: 1299.98 ms Billed Duration: 1300 ms Memory Size: 128 MB Max Memory Used: 98 MB
	No newer events at this moment. Auto retry paused. Resume

Step 6: Configure EventBridge

Created an EventBridge schedule named **DailyS3BucketAudit**.

Configured it as a recurring schedule with a rate of every 5 minutes.

Selected AWS Lambda as the target service.

Configured the target Lambda function as **S3PublicBucketAudit**.

Your schedule DailyS3BucketAudit is being created.

DailyS3BucketAudit

Disable

Edit

Delete

Schedule detail

Schedule name

DailyS3BucketAudit

Description

-

Schedule group name

default

Status

Enabled

Schedule ARN

arn:aws:scheduler:us-east-1:279473426275:schedule/default/DailyS3BucketAudit

Action after completion

NONE

Schedule start time

-

Schedule end time

-

Execution time zone

Asia/Calcutta

Flexible time window

5 minutes

Created date

Jul 23, 2026, 16:57:39 (UTC+05:30)

Last modified date

Jul 23, 2026, 16:57:39 (UTC+05:30)

Schedule

Target

Retry policy

Dead-letter queue

Encryption

Target

Target

S3BucketPublicAudit

Service

AWS Lambda

Payload

-

API

Invoke

Target ARN

arn:aws:lambda:us-east-1:279473426275:function:S3BucketPublicAudit

Execution role

Amazon_EventBridge_Scheduler_LAMBDA_a89fe59bfa

Step 7 :Testing

Run the lamda function and view the CloudWatch Logs

▶	2026-07-23T11:44:39.407Z	No public buckets found.
▶	2026-07-23T11:44:39.427Z	END RequestId: 286a61fe-2f6a-4ed7-ad38-560fa8bb9f5a
▶	2026-07-23T11:44:39.427Z	REPORT RequestId: 286a61fe-2f6a-4ed7-ad38-560fa8bb9f5a Duration: 1339.86 ms Billed Duration: 1340 ms Memory Size: 128 MB Max Memory Used: 98 MB
▶	2026-07-23T11:48:37.688Z	START RequestId: e90a9720-fffa-46a3-9582-7224dd7639ef Version: \$LATEST
▶	2026-07-23T11:48:39.245Z	Public S3 Buckets Found:
▶	2026-07-23T11:48:39.245Z	srikanth-s3-audit-public-test: Block Public Access Disabled, Bucket Policy is Public
▶	2026-07-23T11:48:39.248Z	END RequestId: e90a9720-fffa-46a3-9582-7224dd7639ef
▶	2026-07-23T11:48:39.248Z	REPORT RequestId: e90a9720-fffa-46a3-9582-7224dd7639ef Duration: 1558.35 ms Billed Duration: 1559 ms Memory Size: 128 MB Max Memory Used: 98 MB
		No newer events at this moment. Auto retry paused. Resume

Email Notification

AWS Notifications

<no-reply@sns.amazonaws.com>

to me

▼

5:18 PM (0 minutes ago)

☆

😊

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Public S3 Buckets Found:

srikanth-s3-audit-public-test: Block Public Access Disabled, Bucket Policy is Public

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<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:279473426275:S3PublicBucketAlert:ec91396e-3760-4418-8ba4-7839a543478&Endpoint=srikanth331gugulothu@gmail.com>

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